What the Audits Change: An Interpretable and Fairness-Aware Screening Framework for Diabetes from Population Health Surveys

Abstract— Machine learning models fitted to public health survey data carry failure modes that are invisible in headline accuracy. This paper audits an interpretable screening framework for diabetes on the 2015 Behavioral Risk Factor Surveillance System (BRFSS; n = 253,680, 21 features) and reports what the audits change. Logistic regression, random forest, and histogram-based gradient boosting are compared over five seeds, and three held-out protocols bound duplicate-profile leakage. Leakage is small but not nil: the deduplicated test protocol lowers gradient-boosting AUC from 0.8269 to 0.8177, and 1,963 rows share a covariate vector with a contradicting label. Class-balancing weights cost 60 in calibration (ECE 0.2367 versus 0.0038 unweighted) while buying no discrimination (AUC 0.8265 versus 0.8267); trained unweighted, the model needs no recalibration and reaches slightly higher net benefit. Fairness gaps are quantified with bootstrap intervals and group supports: the sex TPR gap is 0.0171, whereas the age gap of 0.5111 persists at every operating point tested and rises to 0.79 out of time. Attribute-specific thresholds equalize their target attribute in sample but do not transfer: the income gap moves from 0.0899 to 0.8647 by 2021. We therefore report a global sensitivity-targeted threshold, document the mitigation that fails, and state where the data cannot support population-level claims.

Index Terms— diabetes screening, machine learning, calibration, fairness auditing, SHAP, BRFSS, health equity, ethical AI

[conference]IEEEtran cite amsmath,amssymb,amsfonts graphicx textcomp xcolor booktabs url B-.05em i-.025em b-.08em T-.1667em.7exE-.125em X document \\ [Institution] \\ [City], [Country] \\ [email]

Introduction

Problem context and practical requirements

Diabetes is among the most prevalent chronic diseases globally, and a substantial fraction of cases are undiagnosed. Population screening can surface at-risk individuals, but any such tool must be reproducible on open data, transparent in how predictions are formed, honest about what it does not capture, and explicitly bounded against use for individual clinical diagnosis. This paper audits a survey-based screening tool against these requirements. Beyond age and body mass index, diabetes risk reflects blood pressure, cholesterol, physical activity, diet, alcohol use, mental health, and socioeconomic conditions; many of these are self-reported, which makes them accessible but noisy and subject to recall bias, and several are consequences rather than antecedents of diabetes, which limits their use for early-detection reasoning.

Methodological approaches

Prior work has applied logistic regression and, more recently, machine learning classifiers to the BRFSS, with AUC values roughly between 0.72 and 0.80 . Related work has expanded the evidence base in three directions: non-invasive dysglycemia screening on NHANES with calibration and decision-curve analysis ; large-scale external validation of NHANES-trained diabetes models on BRFSS, documenting a 9.7\

Why audits change the conclusions

Reported accuracy is a weak summary of a screening model, and widely used claims about survey-based diabetes models are not reproducible under a leakage-audited protocol: on this same 253,680-row dataset, published pipelines report AUC near 0.99 , whereas we measure 0.827. We therefore audit rather than chase AUC, and three of our findings are negative results that change practice rather than confirm it: class-balancing weights are unnecessary and costly, attribute-specific thresholds do not transfer out of time, and the age disparity is not a threshold artifact.

Research questions and contributions

We address three questions: (1) Driver identification: which features most strongly associate with diabetes status, and how stable are these attributions across seeds and survey waves? (2) Fairness: how do performance and error rates differ across sex, age, income, and education groups, how much of each gap is support starvation, and does any mitigation survive validation? (3) Deployment utility: what training recipe, operating point, and calibration does a screening program actually need? Our contributions are: (i) a three-protocol leakage audit with a reconciled duplicate definition and a label-noise sensitivity analysis; (ii) a calibration audit across four calibrators with per-wave transport analysis, showing class weighting to be an unnecessary burden; (iii) fairness quantification with supports, bootstrap inference, design-effect sensitivity, intersectional cells, and a support-restricted age estimate; (iv) an honest evaluation of attribute-specific thresholds, in sample and out of time, showing they do not transfer; (v) SHAP explanations with stability metrics and comparison against alternative selectors and permutation importance; (vi) out-of-time validation on 2017--2021 waves; and (vii) decision-curve net benefit. The tool is positioned as screening-level and complementing, not replacing, clinical assessment.

Materials and Methods

Data source and scope

We use the CDC Diabetes Health Indicators dataset , derived from the 2015 BRFSS , a telephone survey conducted annually by the CDC since 1984. The cleaned release contains 253,680 listwise-complete respondents and 21 features, under CC0 (public domain). The target is Diabetes_binary (0 = no diabetes, 1 = prediabetes or diabetes; prevalence

13.93\

Variables

The 21 features fall into five groups: clinical (HighBP, HighChol, CholCheck, Stroke, HeartDiseaseorAttack, GenHlth, BMI), lifestyle (Smoker, PhysActivity, Fruits, Veggies, HvyAlcoholConsump), access (AnyHealthcare, NoDocbcCost), mental and physical status (MentHlth, PhysHlth, DiffWalk), and demographic (Sex, Age, Education, Income). Several are proxies---general health status stands in for clinical status, income and education for socioeconomic position---and we treat them as proxies throughout, saying so where results depend on them.

Data preparation and sample flow

Raw: 253,680 respondents no exclusions (no missing values) 253,680 final analytical sample; the ID column was dropped and no row was removed. A stratified 80/20 train/test split (seed 42) yields 202,944 train and 50,736 test cases, with a 64/16/20 train/calibration/test split for calibration work. All preprocessing is fit on the training portion only.

Feature engineering and validation strategy

Features are used as provided (binary or ordinal integers). Continuous-valued fields (BMI, MentHlth, PhysHlth, GenHlth) are standardized for logistic regression only; tree models use raw values. No derived features are added, keeping the screening questionnaire short. Model development uses stratified 5-fold cross-validation on the training portion and a held-out stratified test set for final reporting, with variance reported across five independent seeds (full re-split and re-train) as mean std. We acknowledge the single-split limitation.

Model selection

Three models: (M0) logistic regression with L2 penalty (interpretable baseline), (M1) random forest (50--100 trees, max depth 20), (M2) scikit-learn's histogram-based gradient boosting (150 iterations, 31 leaves). The humble-baseline move: logistic regression is included deliberately and reported honestly. Because class-balancing weights distort the probability scale, we also train an unweighted gradient-boosting variant, and we report both. Baseline settings are fixed and light; to verify that these do not disadvantage any baseline we run a randomized search (20 iterations for gradient boosting, 6 for random forest, 10 for logistic regression; 3-fold CV on the training portion only).

Evaluation framework

We keep the threshold convention explicit, because fairness at a conventional cutoff and fairness at a deployed operating point answer different questions. Unless stated otherwise, subgroup fairness (Table 6) and the support-restricted age analysis (Table 7) use the conventional 0.5 cutoff on raw scores, which is the only convention under which a reader can recompute the confusion matrix; all operating-point results (Table 8) use recalibrated scores at a sensitivity-targeted threshold; and the per-wave thresholds in Table 5 are chosen on raw scores to hold 85\ Discrimination: ROC-AUC, average precision (AUPRC), F1, accuracy. Calibration: Brier score, expected calibration error (ECE), and maximum calibration error (MCE), overall and per group, before and after recalibration with isotonic, Platt (logistic on p), Beta, and Venn--Abers calibrators; ECE is reported under five binning schemes for robustness. Leakage control: three held-out protocols (plain stratified split; deduplicated test set excluding any row whose covariate vector occurs in training; grouped held-out assignment with identical covariate vectors confined to one fold), a full deduplicated refit, and a label-noise refit that removes rows whose covariate vector carries a contradicting label. Fairness: TPR gap, FPR gap, equalized-odds difference , and selection-rate gap, each with per-group supports, 300-resample bootstrap confidence intervals, and sensitivity to assumed design effects; intersections of sex with age, sex with income, and age with income are also reported. Model comparison: paired bootstrap of the test-set AUC difference (2,000 resamples). Utility: decision-curve net benefit across thresholds. Ablations by feature group and by training-only feature ranking.

Explainability method and scenario probes

TreeSHAP for the gradient boosting model on a 2,000-row test sample; mean absolute SHAP per feature; top-10 stability as mean pairwise Jaccard overlap across five seeds and across the 2017--2021 waves. As a complementary view we count single-feature counterfactual flips near the decision threshold, since SHAP is an attribution method and not a causal explanation . Interpretive boundaries: multicollinearity among health indicators, approximation uncertainty, and the absence of any causal interpretation. Separately, bounded perturbation probes shift each modifiable feature to a specified value, clipped to its historical range, and report the induced change in predicted probability with a paired bootstrap confidence interval, on raw and recalibrated scores and broken down by subgroup; these characterise model behaviour and are not projections of causal effects.

Software implementation and reproducibility

Python 3.11.8; scikit-learn 1.5.2 ; XGBoost 2.1.3 ; SHAP 0.46.0; pandas 2.2.3; NumPy 2.1.3; matplotlib 3.9.2. Pinned versions are in requirements.txt; seeds are fixed (42, 7, 123, 2024, 99). Run manifests, per-seed logs, and a SHA-256 hash manifest of every artifact are released with the code .

Results

Predictive performance and model comparison

Table 1 reports test performance over five seeds with AUPRC added, since screening operates under class imbalance. Gradient boosting leads on both metrics (ROC-AUC 0.8288 0.0014, AUPRC 0.4229), and logistic regression trails by less than 0.008 AUC. A paired bootstrap of the test-set AUC difference (2,000 resamples) shows the separations are statistically significant yet practically negligible: HGB exceeds random forest by AUC 0.0034 (95\

TABLE *Test performance (mean std, 5 seeds) and paired AUC comparison.*

Model	Accuracy	F1	ROC-AUC	AUPRC
Logistic Regression	0.7308 0.0013	0.4418 0.0015	0.8212 0.0014	0.3926
Random Forest	0.7458 0.0008	0.4515 0.0007	0.8261 0.0015	0.4153
Hist. Grad. Boosting	0.7203 0.0025	0.4426 0.0023	0.8288 0.0014	0.4229
XGBoost (tuned, single run)	---	---	0.8274	0.4234
5@lPaired bootstrap, HGB minus alternative (95\ HGB - RF	---	---	+0.0034\ (0.0023, 0.0044)	---
HGB - LR	---	---	+0.0073\ (0.0058, 0.0088)	---

Leakage audit and duplicate reconciliation

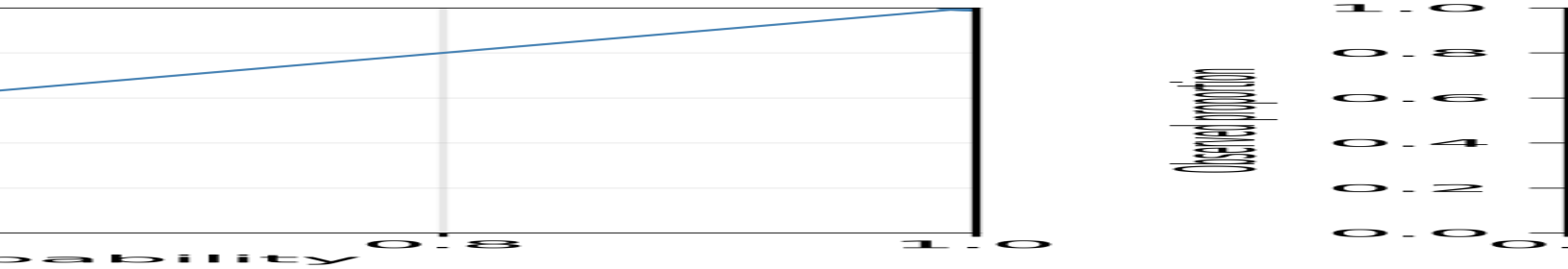
We reconcile two duplicate counts that are frequently conflated. Under a strict definition---rows identical in all columns, including the outcome---24,206 rows (9.54\

TABLE *Three held-out protocols (HGB unless noted).*

Protocol	Test n	ROC-AUC	AUPRC
P1: plain stratified 80/20 --- LR	50,736	0.8196	0.3926
P1: plain stratified 80/20 --- RF	50,736	0.8236	0.4205
P1: plain stratified 80/20 --- HGB	50,736	0.8269	0.4231
P2: deduplicated test --- LR	43,900	0.8096	0.3994

Protocol	Test n	ROC-AUC	AUPRC
P2: deduplicated test --- RF	43,900	0.8143	0.4284
P2: deduplicated test --- HGB	43,900	0.8177	0.4309
P3: grouped held-out (5-fold) --- HGB	253,680	0.8301	0.4326

Calibrator	Brier	ECE	MCE	Mean pred.
Isotonic	0.0976	0.0031	0.2500	0.140
Platt (logistic)	0.0977	0.0083	0.0559	0.139
Beta	0.0974	0.0026	0.0508	0.139
Venn--Abers	0.1266	0.1534	0.2420	0.282
Observed prevalence	---	---	---	0.139



(0.115). Class weighting is thus an avoidable source of calibration error rather than a necessary cost-sensitive choice, and the recalibration stage of this paper is unnecessary for an unweighted model.

TABLE *Class-weighted versus unweighted training (HGB, 2015 test, n=50,736).*

	Class-weighted	Unweighted
ROC-AUC / AUPRC	0.8265 / 0.4229	0.8267 / 0.4234
Raw Brier / ECE	0.1757 / 0.2367	0.0976 / 0.0038
Isotonic ECE / Beta ECE	0.0031 / 0.0026	0.0034 / 0.0042
Mean predicted (observed 0.139)	0.376	0.139
Sensitivity-targeted threshold	0.115	0.115
Sensitivity / specificity at that threshold	0.865 / 0.623	0.857 / 0.637
Net benefit at that threshold	0.0785	0.0787

Calibration audit and post-hoc recalibration

Discrimination alone is insufficient for screening, where a communicated probability must correspond to an observed frequency. Table 4 compares four calibrators fitted on a held-out calibration split. Beta calibration is best on all three metrics (Brier 0.0974, ECE 0.0026, MCE 0.0508), closely followed by isotonic (ECE 0.0031 but MCE 0.2500) and Platt (ECE 0.0083, MCE 0.0559). Venn--Abers performs worst here (ECE 0.1534, mean predicted probability 0.2815). We implement it in its isotonic-plus-prior form: an isotonic regression of the outcome on the base score, combined with a self-consistency correction reconciling the total number of predicted positives with the number of observed positives in the calibration split. That correction assumes the base scores are mildly and roughly uniformly over-confident; a class-weighted ensemble whose mean predicted probability is 2.7 times the true prevalence violates the assumption, and the map shifts probabilities only partially back, leaving ECE at 0.1534. We note this explicitly because the method is usually reported as strong, and the discrepancy is informative rather than anomalous. All calibrators are monotone, so AUC is unchanged (0.8265). Per-group maximum ECE falls from 0.2505 to 0.0045 (sex), 0.3403 to 0.0208 (age), 0.3093 to 0.0343 (income), and 0.3478 to 0.0342 (education) under isotonic recalibration. These conclusions are robust to binning: the uncalibrated ECE is 0.2367 under equal-width binning at 10, 15, and 20 bins, 0.2367 under equal-mass binning, and 0.2155 under threshold-weighted ECE, while the isotonic value stays between 0.0026 and 0.0054 throughout (equal-width 15: 0.0049; equal-mass 10: 0.0044).

TABLE *Calibration before and after post-hoc recalibration (HGB, 2015 test, n=50,736).*

Calibrator	Brier	ECE	MCE	Mean pred.
Uncalibrated	0.1757	0.2367	0.4201	0.376

Fig. *Reliability diagrams before and after isotonic recalibration (HGB, 2015 test).*

Out-of-time validation and calibration transport

Training on the 2015 wave and testing on later waves (Table 5) yields AUC 0.8338 (2017), 0.8256 (2019), and 0.8198 (2021) with a mild downward trend. We also asked whether the 2015 calibration map can be transported rather than refit per wave, since refitting is impossible in a prospective deployment. Applying the 2015-fitted isotonic map to later waves gives ECE 0.0064, 0.0062, and 0.0097 against raw values of 0.2240, 0.2326, and 0.2236; refitting on each wave improves ECE by only 0.0035, 0.0030, and 0.0056 (Brier penalties below 0.002), so the transport penalty is small, and maximum subgroup ECE stays low after transport (age 0.0168--0.0246). The threshold holding 85\

TABLE *Out-of-time validation with transported (2015-fitted) calibration. Wave thresholds are chosen on raw scores to reach 0.85 sensitivity; ECE is on calibrated scores.*

Test wave	n	Prev.	ROC-AUC	Raw ECE	Transport ECE	Thr. (sens.\ 0.85)
2015	50,736	0.139	0.8265	0.2367	0.0031	0.115
2017	275,854	0.141	0.8338	0.2240	0.0064	0.420
2019	243,317	0.145	0.8256	0.2326	0.0062	0.420
2021	236,378	0.142	0.8198	0.2236	0.0097	0.395

Fairness with supports, inference, and design-effect sensitivity

We quantify subgroup disparities with per-group supports and bootstrap confidence intervals (300 resamples) rather than point estimates alone, following recent audit practice for clinical risk models . Table 6 reports the TPR gap, the equalized-odds FPR gap, and the smallest positive count in any group. We caution against over-interpreting gap metrics without base-rate context . The sex TPR gap is small (0.0241, 95\

TABLE *Subgroup disparities with support and bootstrap inference (HGB, single global threshold 0.5 on raw scores).*

Stratifier	TPR gap (95\					
Sex	0.0241 (0.0066, 0.0417)	0.0643	22,205	3,384	0.074	
Age (13 bands)	0.7889 (0.6421, 0.8774)	0.5320	1,207	14	0.407	
Income (8 levels)	0.2849 (0.2571, 0.3173)	0.4229	1,941	475	0.268	
Education (6 levels)	0.2436 (0.2107, 0.2781)	0.4738	795	221	0.330	

Stratifier	TPR gap (95\ Sex	0.0241 (0.0066, 0.0417)	0.0643	22,205	3,384	0.074
Sex Age (21 cells)	0.5078	0.6045	1,200	39	---	
Age Income (64 cells)	0.7778	0.7622	113	31	---	

Support-restricted age disparity, severity stratification, and label noise

Repeating the age computation under increasing minimum-support thresholds (Table 7) shows the gap falls from 0.7879 over all 13 bands to 0.1436 over the seven bands with at least 500 positives, while the widest within-group 95\

TABLE *Age TPR gap under minimum support (single global threshold 0.5), and severity-stratified performance (HGB).*

4c|Age gap vs.\ minimum support	3cSeverity strata						
Min.\ pos.	Bands	TPR gap	Widest CI	Stratum	n	Prev.	AUC
1	13	0.7879	0.340	GenHlth 1--2	26,988	0.057	0.8145
20	12	0.5426	0.340	GenHlth 3	15,064	0.177	0.7443
50	11	0.5426	0.229	GenHlth 4--5	8,684	0.330	0.7199
100	10	0.4206	0.154	BMI <30	33,123	0.092	0.8263
500	7	0.1436	0.072	BMI 30	17,613	0.227	0.7802

Operating points and the failure of attribute-specific thresholds

For screening, sensitivity matters more than overall accuracy. A single global threshold on recalibrated scores, selected to reach 0.85 sensitivity, is 0.115 and yields sensitivity 0.865, specificity 0.623, and net benefit 0.0785. We next implemented the mitigation fairness audits recommend: attribute-specific thresholds, each fitted on the calibration split to deliver 0.85 sensitivity within its own level (sex, age, income, education, or the joint sex-by-age cell). In sample (Table 8), each policy does what it is designed to do---the income policy reduces the income TPR gap to 0.0899, the education policy to 0.0933, the sex policy to 0.0171---and all hold overall sensitivity near 0.85. Out of time, however, the fitted thresholds do not transfer: by 2021 the income policy's own gap has grown to 0.8647 and the education policy's to 0.7128, and the age policy, which should equalize age, produces a TPR gap of 0.9087 in sample and 0.8937--0.8992 out of time---worse than the global threshold. Per-group calibration before thresholding, an alternative stabiliser, narrows the sex gap further (0.0171 0.0072) and the age gap modestly (0.5111 0.4780) but leaves the other attributes unchanged or worse, and it too is fitted on a single split. Diagnosing the failure shows the thresholds are not the problem: between 2015 and 2021 the per-level thresholds drift by at most 0.040, and most are unchanged. What moves is the prevalence within each level, which shifts by up to +0.029 (income level 8) and -0.059 (education level 2) and carries the mean predicted score with it, so the sensitivity each level actually receives drifts between 0.772 and 0.895 rather than the 0.85 targeted. The unevenness of those within-level shifts---not threshold instability---is what widens the group gap. Attribute-specific thresholds fitted on one calibration split therefore do not transfer out of time, and we report the failure rather than a mitigation. What does transfer is the global sensitivity-targeted threshold, which holds sensitivity at 0.84--0.86 and specificity at 0.62--0.64 across all three later waves; but it does not remove the age disparity, whose TPR gap is 0.5111 in 2015 and 0.7727, 0.7901, and 0.7646 in 2017, 2019, and 2021, while the sex gap stays small at 0.0171, 0.0201, 0.0229, and 0.0275. The age gap is a property of the model's score geometry at sensitivity-oriented operating points, not an artifact of threshold choice. A single global threshold also cannot substitute for a validated mitigation: selecting it to maximise specificity subject to overall sensitivity 0.85 and a bound on the worst-group TPR gap shows what a global fairness constraint costs---bounding the gap at 0.50 moves the threshold to 0.400 and specificity to 0.621, bounding it at 0.30 moves it to 0.140 and specificity to 0.344, and bounding it at 0.10 forces the threshold to 0.045 and specificity to 0.147.

TABLE *Threshold policies in sample and out of time. All policies act on recalibrated scores; each level-specific threshold targets 0.85 sensitivity within its level, fitted on the 2015 calibration split.*

Policy	Sens.	Spec.	Sex gap	Income gap	Education gap
6@l2015 test set (in sample)					
Global, sensitivity -targeted	0.865	0.623	0.0171	0.2158	0.1727
Sex-specific	0.865	0.623	0.0171	0.2158	0.1727
Age-specific	0.840	0.634	0.0340	0.2106	0.1876
Income-specific	0.867	0.604	0.0476	0.0899	0.1141
Education-specific	0.859	0.613	0.0288	0.1590	0.0933
Joint sex age	0.846	0.624	0.0144	0.2224	0.1962
6@l2021 wave (thresholds fitted on 2015, transported)					
Global, sensitivity -targeted	0.842	0.639	0.0275	0.3079	0.1411
Sex-specific	0.842	0.639	0.0275	0.3079	0.1411
Age-specific	0.819	0.648	0.0368	0.2987	0.1567
Income-specific	0.734	0.694	0.0461	0.8647	0.2927
Education-specific	0.845	0.619	0.0274	0.2137	0.7128
Joint sex age	0.824	0.637	0.0122	0.3043	0.1566

Explainability findings and stability

SHAP (Fig. 2) ranks general health status (0.638), high blood pressure (0.541), age (0.416), BMI (0.382), and high cholesterol (0.313) as the leading contributors, followed by income (0.136), sex (0.125), cholesterol check (0.092), mental health (0.063), heavy alcohol use (0.060), heart disease (0.059), difficulty walking (0.056), and education (0.045). These align with established risk factors and support the model's face validity, although several are consequences rather than antecedents of diabetes. The top-10 ordering is stable: mean pairwise Jaccard overlap is 0.964 across the five seeds (minimum 0.818, differing only in the tenth position) and 0.818 between 2015 and each later wave, where the substitution is heavy alcohol use for difficulty walking. The ranking is therefore a property of the data period rather than of the seed. As a complementary check on SHAP's known limitations , we compared the SHAP ranking with permutation importance (five repeats, ROC-AUC scoring): the two agree closely (Spearman rank correlation 0.955; 9 of the top 10 features in common), differing only in the order of BMI and age and in the tenth position. We also counted single-feature counterfactual flips among the 4,408 individuals whose score lies within 0.05 of the threshold: a one-point change in general health flips 67.5\

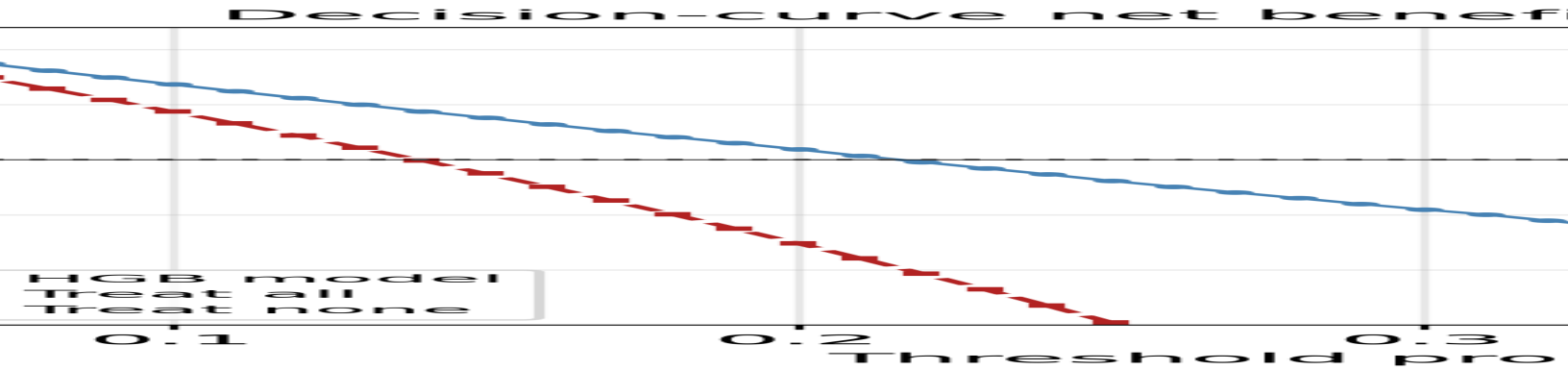


Fig. *Top 12 features by mean |SHAP| (HGB, 2,000-row test sample).*

Fig. *Decision-curve net benefit versus threshold probability (HGB, 2015 test).*

Feature subset analyses

Table 9 combines the feature-group ablations with the minimal-questionnaire experiment and a comparison of ranking procedures. Clinical-history features carry most of the signal (AUC 0.8241 versus 0.7104 for demographics alone), while lifestyle features alone are weaker (0.7475). Using training-only SHAP rankings, five features retain AUC 0.8186 against 0.8269 for all 21, and eleven reach 0.8257. Comparing three training-only ranking procedures at k=10 against a random-subset baseline, SHAP attains 0.8248, correlation filtering 0.8218, mutual information 0.8213, and ten random features 0.7784 0.0108, so SHAP is strongest but only marginally over the other filters. For context, a copula tail-dependence filter on this dataset has been reported at 0.823 with 10 features , consistent with our range. Because several top features proxy the same constructs, we do not claim that a ten-item form is clinically validated.

TABLE *Feature subset analyses (HGB, 2015 test).*

4c|Feature-group ablation				3cSubset and selector comparison			
Feature group	n	ROC-AUC	F1	Setting	n	ROC-AUC	F1
Demographic only	4	0.7104	---	Top-5 SHAP	5	0.8186	0.4305
+ clinical history	11	0.8241	---	Top-8 SHAP	8	0.8231	0.4357
+ lifestyle	14	0.7475	---	Top-11 SHAP	11	0.8257	0.4404
Full	21	0.8269	0.4411	All features	21	0.8269	0.4411
				Mutual info k=10	10	0.8213	---
				Correlation k=10	10	0.8218	---
				Random k=10	10	2c0.7784 0.0108	

Decision-curve utility

For a screening program the operating point must be justified by net benefit rather than by a generic accuracy target. Figure 3 gives decision-curve net benefit against threshold probability: the model dominates treat-all and treat-none across the clinically relevant range, with net benefit 0.068 against 0.044 for treat-all at a threshold of 0.10 and 0.009 against -0.076 at 0.20, treat-all becoming actively harmful beyond 0.10. The curve therefore supports the sensitivity-oriented operating points of Table 8 and indicates a useful threshold range well below 0.5, consistent with the strong over-prediction of the class-weighted model.

Scenario probes

We performed bounded perturbation probes on the test set, clipping each modified feature to its historical range; these are sensitivity probes of model behaviour, not projections of causal effects. A +5 BMI shift raises mean predicted probability by 0.0694 on raw scores but only 0.0410 on recalibrated scores, and a one-point worsening of general health by 0.0985 raw and 0.0499 recalibrated: recalibration compresses the response, so probe magnitudes should be quoted on the scale of the scores actually deployed. The smoking and physical-activity probes are small and not distinguishable from zero (smoker 01: +0.0002, CI [-0.0208, 0.0128]; physical activity 10: +0.0036, CI [-0.0042, 0.0230]), and setting heavy alcohol use to present decreases mean predicted probability by 0.1034 (CI [-0.3094, 0.0031]), with only 4.4\

Discussion

What the audits changed

The audit work materially altered six claims. First, the age TPR gap is not a single number: it is 0.7889 over all 13 bands but 0.1436 over the seven bands with at least 500 positives, and it is not removed by threshold choice. Second, class-balancing weights are unnecessary: they buy no discrimination, cost a factor of 62 in ECE, and are better replaced by a sensitivity-targeted threshold on an unweighted model. Third, the AUC differences between models, though statistically significant, are too small to motivate a more complex model. Fourth, the headline AUC is protocol-dependent: the leakage-free estimate is 0.8177, not 0.8269, and published figures near 0.99 are not reproducible under any of our three protocols. Fifth, the mitigation fairness audits recommend---attribute-specific thresholds---equalizes its target attribute in sample but does not transfer out of time, and we report the failure rather than a remedy. Sixth, the top features are stable across seeds and waves, but counterfactual flip counts show that SHAP magnitude does not correspond to decision leverage for binary features.

Contextual positioning within the literature

Our discrimination (AUC 0.8288, AUPRC 0.4229) exceeds earlier BRFSS-based studies reporting 0.72--0.80 and is close to NHANES-based non-invasive dysglycemia screening at AUC 0.820 , but we report AUPRC and decision-curve net benefit as well, since discrimination alone is not comparable across studies with different label definitions and prevalence. It is markedly more conservative than BRFSS studies reporting AUC near 0.99 with stacked ensembles and resampling-based balancing ; our leakage audit indicates that on this same 253,680-row dataset, duplicate profiles and fold construction move AUC by at most 0.010, so we regard those figures as unlikely to be reproducible. Where an NHANES-trained model was externally validated on BRFSS, AUC fell from 0.794 to 0.717 under distribution shift ; our in-domain 2015-to-later-wave transfer is milder (0.8338 to 0.8198), consistent with the absence of a cross-survey shift. For feature reduction, a copula tail-dependence filter on this same dataset reports AUC 0.823 with 10

features , against our 0.8248 for SHAP at k=10; direct comparison remains limited by differing years, features, splits, and label definitions, and we do not claim a decisive win. On calibration, our finding that class weighting is the dominant source of miscalibration is consistent with the wider literature, in which isotonic and Platt can themselves degrade proper scoring for strong tabular learners ; we therefore compare four calibrators rather than assuming one. Our ranking agrees in two respects---Beta calibration is the most consistently beneficial, and isotonic shows the highest maximum calibration error (0.2500 here) even as its expected calibration error is small---and differs in one, which we attribute to our base model: Venn--Abers performs worst here but is strongest in that benchmark, because its self-consistent prior correction assumes a mildly and uniformly over-confident base model, which a class-weighted ensemble with mean predicted probability 2.7 times the true prevalence is not. A calibrator must be matched to the base model's miscalibration mode, not chosen by a general ranking.

Practical deployment scenarios

(1) Population-level risk stratification for outreach, using an unweighted model with a sensitivity-targeted threshold and no recalibration requirement; (2) questionnaire triage in resource-limited settings, using the 8--10 item reduced form; (3) hypothesis generation for public-health research. Each requires corroboration with clinical measurements before any individual decision. Because our only validated operating point is a single global threshold, we do not recommend subgroup thresholds on this evidence.

Methodological limitations and validity boundaries

Numbered: (1) self-reported cross-sectional data, no biomarkers (HbA1c, glucose), and several leading predictors are consequences rather than antecedents; (2) race/ethnicity absent from this release, limiting health-equity analysis and precluding race-stratified reliability; (3) no survey design weights, so population-level representativeness is not established and our intervals are optimistic by an unquantified design effect; (4) prediabetes and diabetes are collapsed into one label, so the severity-stratified analysis is a proxy rather than a substitute; (5) the leakage-free estimate uses a deduplicated test set of 43,900 rows, calibration was restored post hoc on a single 2015 split (the 2023 wave lacks four features and was excluded), and our out-of-time validation remains within BRFSS, so no cross-survey external validation is attempted---a genuine gap, since label definitions and measurement differ between BRFSS and NHANES and prior work shows AUC falling from 0.794 to 0.717 under that shift ; (6) the randomized search budget is small and single-split; (7) the attribute-specific threshold policies failed out-of-time validation, so no fairness mitigation is validated here, and intersectional cells are descriptive, with the smallest cell holding 31 positives; (8) SHAP is an attribution method, and no formal faithfulness guarantee is offered. Throughout, association is not causation and screening is not diagnosis. Design-aware modelling and design-based inference are the established remedy for population-level claims , and we identify them as prerequisites for any policy-facing use. The magnitude of the issue is not hypothetical: national BRFSS estimates are produced by raking to a dozen margin sets (sex by age, race/ethnicity, education, marital status, home ownership, phone usage, and state), and the CDC's own analysis of aggregated state weights documents a concrete national-distribution error of 0.5 percentage points on a single demographic margin, with downstream distortion of diabetes prevalence estimates by demographic group . The cleaned release we use carries none of those variables, so the design-aware route requires the raw BRFSS XPT files and a re-derivation of the respondent-level merge, which we have not attempted. Future work: incorporation of BRFSS design weights with design-based uncertainty quantification ; race/ethnicity from the raw BRFSS files with race-stratified reliability; a fairness mitigation validated out of time before deployment ; extension to the 2023 wave and to cross-survey validation against NHANES ; and prospective evaluation with clinical biomarkers.

Conclusions

We audited an interpretable, fairness-aware screening framework for diabetes on 253,680 BRFSS respondents. (1) Gradient boosting and random forest outperform logistic regression by a statistically significant but practically negligible margin (AUC 0.0073). (2) The headline AUC is protocol-dependent: the leakage-free estimate is 0.8177, and 1,963 rows carry a covariate vector with a contradicting label, whose removal would raise AUC to 0.8397. (3) Class-balancing weights are unnecessary and costly (ECE 0.2367 versus 0.0038), and an unweighted model with a sensitivity-targeted threshold gives slightly higher net benefit. (4) The sex TPR gap is small (0.0171 at the deployed operating point), while the age gap persists at every operating point tested, is only partly support-starved, and rises out of time. (5) Attribute-specific thresholds---the mitigation fairness audits recommend---do not transfer out of time, and we report that failure rather than claim a remedy. (6) Decision-curve net benefit supports sensitivity-oriented operating points below 0.5. The tool is a screening-level, hypothesis-generating instrument for population triage and research, not a validated diagnostic or causal instrument, and it does not yet support population-level prevalence claims.

Acknowledgments

The authors thank the CDC for the BRFSS data and the UCI repository for hosting.

Data Availability

The dataset is publicly available . All code, run manifests, logged results, and a SHA-256 hash manifest of every artifact are released at .

Author Contributions

[TBD: CRediT statement.]

Conflicts of Interest

The authors declare no conflict of interest.

References

[1] CDC Diabetes Health Indicators [Dataset]. (2017). UCI Machine Learning Repository. https://doi.org/10.24432/C53919 (accessed on 17 Oct. 2026).

[2] Z. Xie, O. Nikolayeva, J. Luo, and D. Li, ``Building risk prediction models for type 2 diabetes using machine learning techniques," Prev. Chronic Dis., vol. 16, p. e130, 2019. https://doi.org/10.5888/pcd16.190109.

[3] S. M. Lundberg and S.-I. Lee, ``A unified approach to interpreting model predictions," in Adv. Neural Inf. Process. Syst., vol. 30, pp. 4768--4777, 2017.

[4] M. Hardt, E. Price, and N. Srebro, ``Equality of opportunity in supervised learning," in Adv. Neural Inf. Process. Syst., vol. 29, pp. 3315--3323, 2016.

[5] N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman, and A. Galstyan, ``A survey on bias and fairness in machine learning," ACM Comput. Surv., vol. 54, no. 6, art. 115, 2021. https://doi.org/10.1145/3457607.

[6] B. Sun, C. Zhang, K. Ji, and X. Lu, ``Beyond the blood draw: Explainable machine learning for non-invasive dysglycemia risk screening," arXiv:2606.16056, 2026.

[7] R. S. Pall, S. Yadav, S. Bhalerao, S. Sahu, R. Ahluwalia, and B. Awadhiya, ``Comprehensive evaluation of machine learning for type 2 diabetes risk prediction: Large-scale external validation and fairness analysis," in Proc. IEEE CIPHER, 2026. https://doi.org/10.1109/CIPHER70417.2026.11523789.

[8] A. Aich, M. M. Murshed, S. Hewage, and A. Mayeaux, ``A copula based supervised filter for feature selection in machine learning driven diabetes risk prediction," Sci. Rep., 2026. https://doi.org/10.1038/s41598-026-41874-9.

[9] M. Matabuena, J. C. Vidal, R. Ghosal, and J.-P. Onnela, ``Screening for diabetes mellitus in the U.S. population using neural network models and complex survey designs,'' arXiv:2403.19752, 2024.

[10] S. Roy, S. Girmachew, and N. Chavan, ``KAISEN: Reproducible subgroup fairness auditing for clinical risk models,'' arXiv:2607.28608, 2026.

[11] J. Wieczorek, ``Design-based conformal prediction,'' Survey Methodology, vol. 49, no. 2, pp. 305--322, 2023.

[12] Centers for Disease Control and Prevention, ``National weighting of data from the Behavioral Risk Factor Surveillance System (BRFSS),'' 2016. https://stacks.cdc.gov/view/cdc/43284 (accessed on 21 Oct. 2026).

[13] Centers for Disease Control and Prevention, ``Behavioral Risk Factor Surveillance System,'' 2015 annual data. https://www.cdc.gov/brfss/annual_data/annual_2014.html (accessed on 17 Oct. 2026).

[14] F. Pedregola et al., ``Scikit-learn: machine learning in Python,'' J. Mach. Learn. Res., vol. 12, pp. 2825--2830, 2011.

[15] T. Chen and C. Guestrin, ``XGBoost: a scalable tree boosting system,'' in Proc. ACM SIGKDD, pp. 785--794, 2016.

[16] O. L\'etoff\'e, X. Huang, and J. Marques-Silva, ``Towards rigorous explainability by feature attribution,'' arXiv:2604.15898, 2026.

[17] V. Manokhin and D. Grnhaug, ``When post-hoc calibration hurts: Platt scaling and isotonic regression on modern tabular models,'' in Proc. Symp. Conformal and Probabilistic Prediction with Applications, ser. PMLR, vol. 329, pp. 378--381, 2026.

[18] V. Manokhin and D. Grnhaug, ``Classifier calibration at scale: An empirical study of model-agnostic post-hoc methods,'' arXiv:2601.19944, 2026.

[19] A. Author, ``BRFSS diabetes screening: code, run manifests, and SHA-256 artifact manifest,'' GitHub repository. https://github.com/Aldrin7/brfss-diabetes-screening-iccica2027 (accessed on 21 Oct. 2026). thebibliography